

Queries (you can use any web tool however we recommend those mentioned in the laboratory)

You must send the file containing your exam answers to this e-mail address:
`curs.la.nlp@fmi.unibuc.ro`

The exam answers must be sent only from the student's institutional e-mail address.

The phrases table file (for exercises 1 and 3):

https://docs.google.com/spreadsheets/d/1c1YNI9Y2rb4IIRoW2gqoVfAtv9KE_A4W78J9KTDczQM/edit?usp=sharing

Each student uses the phrase that appears next to his/her name in the table

1. Well formed substring tables (1p)

Perform a syntactic analysis for the given input string (you can find it in the phrases table) using the well formed substring tables algorithm.

Create a minimal CFG grammar (it should contain only the rules that are necessary in the analysis) and explain the meaning of the non terminal symbols (Ex: VP= verb phrase)

How to solve: Draw the parse graph for the given phrase. Add numbers to the graph connections (the numbers must indicate the order in which the oriented edges appear). For each oriented edge also write the production rule responsible for it.

2. Text preprocessing (1p)

Explain the difference between stemming and lemmatization. What is the role of these operations? Why do we use them?

3. Dependency parsing (1p)

Perform dependency parsing using the Stanford parser for the given input string (found in the phrases table).

Draw the resulting graph.

4. WordNet++ (1p)

How is the disambiguation context of a Wikipage created?

5. Naive Bayes disambiguation (1p)

Present a way of performing feature selection (of semantic or of syntactic type) for the Naive Bayes model used in local unsupervised WSD.

6. Ant Colony Algorithm (2p)

- a) What is the purpose of the odor vectors? Find two cases in which the odor vector doesn't change for a node.
- b) What are bridges and what is their role in the algorithm?

7. Shotgun WSD

- a) What does a sense embedding mean? What role does it play in the Shotgun WSD algorithm? Describe the method used to compute a sense embedding?
- b) When do we combine overlapping sense reads? What is the role of combining overlapping sense reads? Describe the method of computing overlapping sense reads.

1. (1p) Enumerați și explicați cel puțin 3 operații diferite care pot face parte din preprocesarea unui text. / List and explain at least 3 operations which can constitute steps in text preprocessing.
2. (2p) În cazul analizei sintactice bazate pe constituenți, dați un exemplu de algoritm top-down și aplicați-l pe fraza următoare:
I did all my homework and spent the entire week learning for the exam.
 Pentru aplicarea algoritmului concepeți o gramatică CFG minimală (fără reguli în plus) și explicați semnificația notatiilor folosite pentru neterminale (Ex.: VP = grup verbal).

 For constituent based syntactic parsing, give an example of a top-down algorithm. Use it to parse the following sentence.
I did all my homework and spent the entire week learning for the exam.
 Create a minimal CFG grammar (it should contain only the rules that are necessary in the analysis) and explain the meaning of the non terminal symbols (Ex: VP= verb phrase)
3. (1p) Menționați două niveluri la care poate apărea ambiguitatea în text; dați câte un exemplu (câte o propoziție) relevant pentru fiecare tip de ambiguitate, și explicați de unde provine ambiguitatea. / Mention two linguistic levels where ambiguity can occur, and give one example sentence which illustrates each type of ambiguity; explain where the ambiguity arises from.
4. (1p) Explicați diferența între modelul skip-gram și modelul CBOW pentru antrenarea word embeddings, și descrieți avantajele fiecăruia. / Explain the difference between the skip-gram and the CBOW model for training word embedding representations, and briefly describe the advantages of each method.
5. (1p) Menționați relațiile din WordNet considerate ca fiind cele mai relevante în cazul substantivelor, adjectivelor și verbelor și explicați semnificația lor. / Specify and explain the WordNet relations which are considered to be the most relevant in the case of nouns, adjectives and verbs.
6. (1p) Explicați cum se calculează scorul de înrudire bazat pe sense embeddings între două synseturi în cazul algoritmului Shotgun de dezambiguizare. / Explain how to compute the relatedness score between two synsets based on sense embeddings in the Shotgun disambiguation algorithm.
7. (1p) Ce reprezintă cuiburile în algoritmul Ant Colony folosit pentru dezambiguizarea sensului cuvintelor? / What do nests represent in the Ant Colony algorithm for word sense disambiguation?
8. (2p) Prezentați o modalitate de a face selecția caracteristicilor sintactice în cazul dezambiguizării locale nesupervizate care folosește modelul Bayes naiv. Explicați presupunerea pe care se bazează algoritmul Bayes naiv - care îl face "naiv" - și menționați modelul folosit de algoritm. / Present a way of performing feature selection of syntactic type for the Naive Bayes model used in local unsupervised WSD. Explain